Training Course on DoE

DoE for dependent observations

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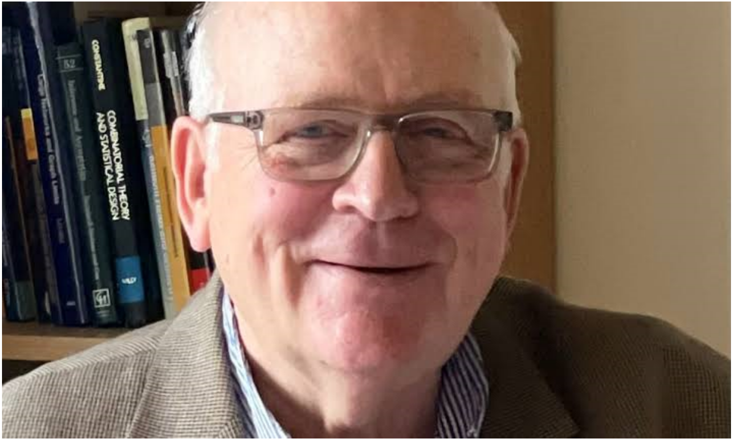
JOHANNES KEPLER UNIVERSITY LINZ

Drongen, June 14, 2026

mODa 14
Model-Oriented Data Analysis and Optimum Design

(dedicated to the memory of H.P. Wynn)

Henry Philip Wynn (1945-2024)



Outline of the presentation

- Connecting to the introduction to (Optimal) Design of Experiments
- Design for Copula models
- Design for Gaussian Processes
  - Estimation
  - Prediction
- Software demonstration

The formula that killed Wall Street (in 2008)

see Salmon, Felix (March 2009). "Recipe for Disaster: The Formula That Killed Wall Street". *Wired Magazine* 17 (3).

$$P[T_A < 1, T_B < 1] = \Phi_2(\Phi^{-1}(F_A(1)), \Phi^{-1}(F_B(1)), \rho)$$

David X. Li's Gaussian copula applied to (joint) default probabilities of mortgages.

"For me, this [blaming Li's formula] is akin to blaming Einstein's $E = mc^2$ formula for the destruction wreaked by the atomic bomb.", P. Embrechts (ETH Zürich), 2009.

The Gaussian copula gives asymptotic independence, i.e. irrespective of how high the correlation is, extreme events occur independently in each margin.

A graphical explanation

from Embrechts, Paul (2009), "Did a Mathematical Formula Really Blow up Wall Street?" *Presentation Slides*.

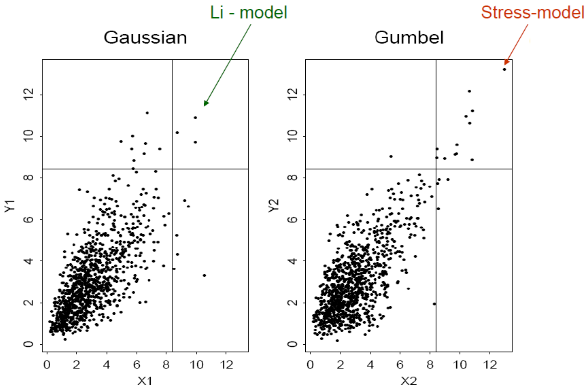


FIGURE 1. 1000 random variates from two distributions with identical Gamma(3,1) marginal distributions and identical correlation $\rho = 0.7$, but *different* dependence structures.

What is a copula function good for?)

R. T. Clemen, T. Reilly, Correlations and copulas for decision and risk analysis, *Management Science* 45(2) : 208–224 (1999).

Definition (Copula)

Let $\mathbb{I} = [0,1]$. A *two-dimensional copula* (or *2-copula*) is a bivariate function $\mathrm{C} : \mathbb{I} \times \mathbb{I} \longrightarrow \mathbb{I}$ with the following properties:

1. for every u, $v \in \mathbb{I}$

$$\mathrm{C}(u,0) = 0,\ \mathrm{C}(u,1) = u,\ \mathrm{C}(0,v) = 0,\ \mathrm{C}(1,v) = v;$$

2. for every u_1, v_1, u_2, $v_2 \in \mathbb{I}$ such that $u_1 \le u_2$ and $v_1 \le v_2$,

$$\mathrm{C}(u_2,v_2) - \mathrm{C}(u_2,v_1) - \mathrm{C}(u_1,v_2) + \mathrm{C}(u_1,v_1) \ge 0.$$

Sklar's Theorem

bivariate case

Theorem (Sklar's Theorem)

Let $F_{Y_1Y_2}$ be a joint distribution function with marginals F_{Y_1} and F_{Y_2}. Then there exists a 2-copula C such that

$$F_{Y_1Y_2}(y_1,y_2) = \mathrm{C}(F_{Y_1}(y_1), F_{Y_2}(y_2))$$

for all reals y_1, y_2.

If F_{Y_1} and F_{Y_2} are continuous, then C is unique.

Conversely, if C is a 2-copula and F_{Y_1} and F_{Y_2} are distribution functions, then the function $F_{Y_1Y_2}$ given by (2) is a joint distribution with marginals F_{Y_1} and F_{Y_2}.

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A regression model

based on (bivariate) copulas

Let us consider a vector $x^T = (x_1, \ldots, x_r) \in X$ of control variables, where $X \subset \mathbb{R}^r$ is a compact set.
Observations come in pairs:

$$y(x) = (y_1(x), y_2(x)),$$

with

$$E[Y(x)] = \eta(x, \beta) = (\eta_1(x, \beta), \eta_2(x, \beta)),$$

where $\beta = (\beta_1, \ldots, \beta_k)$ is a certain unknown parameter vector to be estimated and η_i are known functions.

Define $c_Y(y(x, \beta), \alpha)$ the joint probability density function of the random vector Y, where $\alpha = (\alpha_1, \ldots, \alpha_l)$ are unknown parameters.

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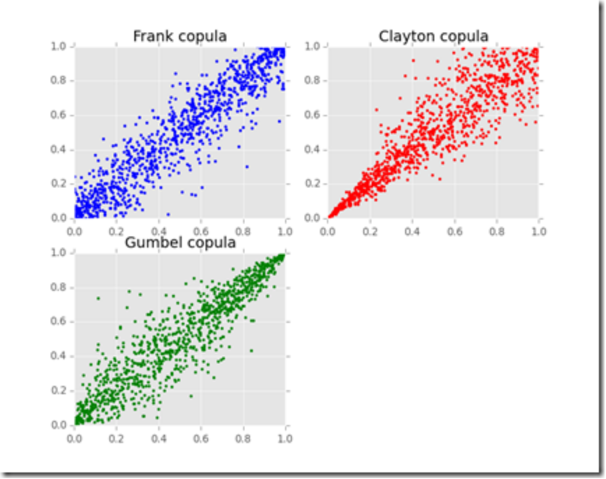
Scatterplots of typical copulas

from the internet



Used copula functions

connect through Kendall's τ

1. *Product Copula,* which represents the independence case:

$$C(u_1, u_2) = u_1 u_2,$$

with $\tau = 0$.

2. *Clayton Copula:*

$$C_\alpha(u_1, u_2) = \left[\max\left(u_1^{-\alpha} + u_2^{-\alpha} - 1,\, 0\right)\right]^{-\frac{1}{\alpha}},$$

with $\alpha \in (0, +\infty)$ and $\tau = \frac{\alpha}{\alpha+2}$.

3. *Gumbel Copula:*

$$C_\alpha(u_1, u_2) = \exp\left(-\left[(-\ln u_1)^\alpha + (-\ln u_2)^\alpha\right]^{\frac{1}{\alpha}}\right),$$

with $\alpha \in [1, +\infty)$ and $\tau = \frac{\alpha-1}{\alpha}$.

The approach

completing the model

Remember

according to Sklar's theorem the joint probability density function is the density of the copula function such that

$$F_{Y_1,Y_2}(y_1,y_2;\alpha) = \int c_Y(y(x,\beta),\alpha)dy =$$

$$= C(F_{Y_1}(y_1),F_{Y_2}(y_2);\alpha)$$

For r independent observations at $x_1,\ldots,x_r$, the corresponding Information matrix is

$$M(\xi,\beta,\alpha) = \sum_{i=1}^{r} w_i J(x_i,\beta,\alpha)$$

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The Fisher information

Definition

For a single observation the FI matrix is a $(k+l)\times(k+l)$:

$$J(x,\beta,\alpha) = \begin{pmatrix} J_{\beta\beta}(x) & J_{\beta\alpha}(x) \\ J_{\beta\alpha}^T(x) & J_{\alpha\alpha}(x) \end{pmatrix}$$

where $J_{\beta\beta}(x)$ is the $(k\times k)$ matrix with the (i,j)th element defined as

$$E\left(-\frac{\partial^2}{\partial\beta_i\partial\beta_j}\log c_Y(y(x,\beta),\alpha)\right) =$$

$$= E\left(\left(\frac{\partial}{\partial\beta}\log c_Y(y(x,\beta),\alpha)\right)\left(\frac{\partial}{\partial\beta}\log c_Y(y(x,\beta),\alpha)\right)^T\right)$$

Optimal design

recap of basics

Design

$$\xi = \left\{ \begin{array}{cccc} x_1 & x_2 & \dots & x_n \\ w_1 & w_2 & \dots & w_n \end{array} \right\},$$

where $\sum_{i=1}^{r} w_i = 1$.

Aim

We are concerned with finding $\xi^*(\beta,\alpha)$ such that maximizes some scalar function $\phi(M(\xi,\beta,\alpha))$.

D-optimality

For now we consider the function $\phi(M) = \log\det M$, i.e. *D-optimality*.

Optimal design

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$$\xi = \begin{Bmatrix} x_1 & x_2 & \dots & x_n \\ w_1 & w_2 & \dots & w_n \end{Bmatrix},$$

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Elisa Perrone

phD student at our department 2012-2015, now at TU Eindhoven

E. Perrone and W. G. Müller, Optimal Designs for Copula Models, *Statistics* **50**(4) : 917–929 (2016).

Design theory

Kiefer and Wolfowitz type, cf. M. A. Heise and R. H. Myers, Optimal Design for Bivariate Logistic Regression, *Biometrics*, 52(2) : 613–624 (1996).

Theorem (Equivalence Theorem: D-criterion)

For a localized parameter vector $(\bar{\beta}, \bar{\alpha})$, the following properties are equivalent:

1. ξ^* *is D-optimal;*
2. $\text{tr}\,[M(\xi^*, \bar{\beta}, \bar{\alpha})^{-1} J(x, \bar{\beta}, \bar{\alpha})] \leq (k+l)$, $\forall x \in X$;
3. ξ^* *minimize* $\max_{x \in X} \text{tr}\,[M(\xi^*, \bar{\beta}, \bar{\alpha})^{-1} J(x, \bar{\beta}, \bar{\alpha})]$, *over all* $\xi \in \Xi$.

The function $d(x, \xi^*) = \text{tr}\,[M(\xi^*, \bar{\beta}, \bar{\alpha})^{-1} J(x, \bar{\beta}, \bar{\alpha})]$ is called the *sensitivity function* for the D-criterion.

Two designs ξ, ξ^* can be compared by the following ratio (*D-Efficiency*):

$$\left(\frac{|M(\xi, \beta, \alpha)|}{|M(\xi^*, \beta, \alpha)|} \right)^{1/(k+l)}.$$

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A first (classical) example

from V. V. Fedorov, The design of experiments in the multiresponse case, *Theory of Probability and its Applications*, 16(2) : 323–332 (1971).

The model

Let us consider two random variables Y_1 and Y_2 such that:

$$E[y_1(x)] = \beta_0 + \beta_1 x + \beta_2 x^2; \;\; E[y_2(x)] = \beta_3 x + \beta_4 x^3 + \beta_5 x^4$$

for each observation x, $0 \leq x \leq 1$. In the original formulation Y_1 and Y_2 are independent with Gaussian margins.

Our investigations

How do different dependence structures affect the optimal designs?

The joint distribution is, then

$$F_Y(y_1, y_2) = C(\Phi(y_1 - \eta_1(x,\beta)), \Phi(y_2 - \eta_2(x,\beta)); \alpha)$$

where C is respectively say the Gumbel, Clayton and Frank copula.

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Plots

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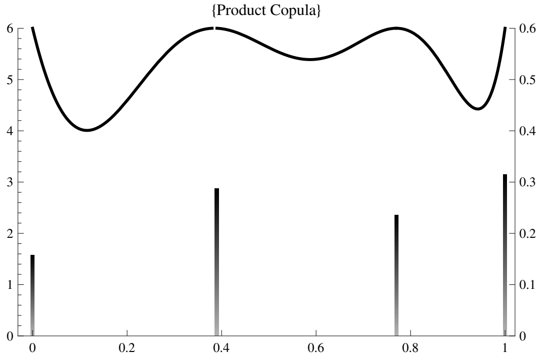

A classical example

Results and comments

	FGM		Clayton		Frank	
τ	α	D-eff	α	D-eff	α	D-eff
-0.10	-0.45	0.23	n.d.	-	-0.90	0.10
-0.05	-0.22	0.59	n.d.	-	-0.45	0.10
0.05	0.22	0.68	0.10	0.16	0.45	0.10
0.10	0.45	0.39	0.22	0.13	0.90	0.10
0.35	n.d.	-	1.08	0.11	3.51	0.11
0.75	n.d.	-	6.00	0.27	14.13	0.16

Table: Losses in D-efficiency by ignoring the dependence in percent.

A clinical trial application

cf. N. G. Denman, J. M. McGree, J.A. Eccleston, S.B. Duffull, Design of experiments for bivariate binary responses modelled by Copula functions, *Computational Statistics and Data Analysis*, 55(4) : 1509–1520 (2011).

Bivariate binary model

Let Y_1 and Y_2 be two random variables that model toxicity and efficacy. For each observation there are four possible outcomes $\{(0,0),(0,1),(1,0),(1,1)\}$.

The marginal probabilities of success are given by:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_{i0} + \beta_{i1}x \text{ with } i = 1,2 \tag{1}$$

where $x \in [0,10]$ and the initial $\beta_1 = [-1,1]$ and $\beta_2 = [-2,0.5]$.

A common choice for the joint distribution function is the standard Gumbel cdf (see Heise and Meyers, 1996) which corresponds to the FGM copula: $p_{11} = F_{Y_1,Y_2}(\pi_1,\pi_2) = \pi_1\pi_2[1+\alpha(1-\pi_1)(1-\pi_2)]$.

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Example - Bivariate binary model

Generalities

The remaining probabilities can be derived as:

$$p_{10} = \pi_1 - p_{11}, \quad p_{01} = \pi_2 - p_{11}, \quad p_{00} = 1 - \pi_1 - \pi_2 + p_{11}.$$

For this model the Fisher information Matrix for a single observation can be calculated as follows (see Dragalin and Fedorov, 2006):

$$M(\theta, \xi_i) = \frac{\partial p}{\partial \theta}\left(P^{-1} + \frac{1}{1 - p_{11} - p_{10} - p_{01}} ee^T\right)\frac{\partial p}{\partial \theta} \qquad (2)$$

where $p = (p_{11}, p_{10}, p_{01})$, $P = diag(p)$ and $e = (1,1,1)^T$.

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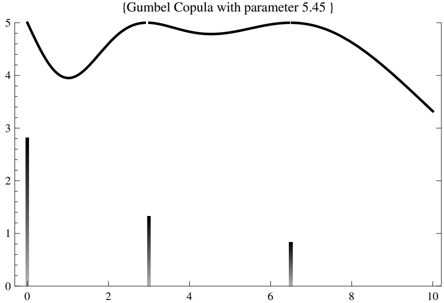
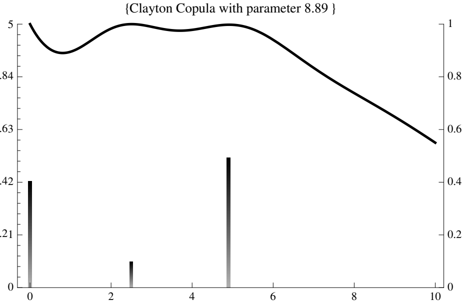
Plots

from Perrone and Müller (2016), $\tau = 0.7$.





Sensitivity analysis

from Perrone and Müller (2016).

1. We find the D-optimal design for the Gumbel model by fixing as localized parameter $\alpha = 0$ (case of independence).
2. We change the structure of the dependence and in particular we enlarge the range of the association by using the Frank, Clayton, and Gumbel copula family comparing the results for the same τ level.

	Frank		Clayton		Gumbel	
τ	α	D-eff	α	D-eff	α	D-eff
0.11	1.00	0.01	0.24	2.42	1.12	0.87
0.45	5.00	0.36	1.68	1.13	1.84	0.50
0.66	10.00	3.18	3.98	1.34	3.00	2.84
0.76	15.00	5.63	6.42	5.54	4.21	5.13
0.82	20.00	6.24	8.89	13.94	5.45	6.12

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0.66	10.00	3.18	3.98	1.34	3.00	2.84
0.76	15.00	5.63	6.42	5.54	4.21	5.13
0.82	20.00	6.24	8.89	13.94	5.45	6.12

Results

Comparisons by assuming different dependence structures.

True Copula	Frank		Clayton		Gumbel	
Assumed Copula	Clayton	Gumbel	Frank	Gumbel	Frank	Clay
$\tau = 0.11$	2.24	0.67	1.99	2.70	0.82	2
$\tau = 0.45$	0.26	0.03	0.26	0.11	0.03	0
$\tau = 0.66$	1.09	0.11	1.04	1.28	0.14	1
$\tau = 0.76$	4.27	0.02	3.87	4.08	0.01	4
$\tau = 0.82$	8.24	0.01	10.91	10.96	0.01	8

Table: Losses in D-efficiency in percent.

Comments

There are significant losses in D-efficiency by assuming different values of the association between the random variables as well as for the same τ and different dependence structures.

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A more flexible model

Model construction

Idea

Considering the strength of the dependence itself dependent upon the regressors x by modeling the τ by a logistic as follows:

$$\boxed{\tau(x,\alpha_1) = \frac{e^{\alpha_1 x - c}}{1 + e^{\alpha_1 x - c}},}$$

where c is a constant chosen such that τ takes values in $[\varepsilon, 1]$ for $\alpha_1 \in [0,1]$ (here $\varepsilon = 0.05$).

How to model the dependence

We model p_{11} by a convex combination of the Clayton and the Gumbel copulas by linking them at the same τ values:

$$\boxed{C(\pi_1,\pi_2;\alpha_1,\alpha_2) = \alpha_2 C_1(\pi_1,\pi_2;2\,e^{\alpha_1 x - c}) + (1-\alpha_2) C_2(\pi_1,\pi_2;1+e^{\alpha_1 x - c}).}$$

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where c is a constant chosen such that τ takes values in $[\varepsilon, 1]$ for $\alpha_1 \in [0,1]$ (here $\varepsilon = 0.05$).

How to model the dependence

We model p_{11} by a convex combination of the Clayton and the Gumbel copulas by linking them at the same τ values:

$$\boxed{C(\pi_1,\pi_2;\alpha_1,\alpha_2) = \alpha_2 C_1(\pi_1,\pi_2; 2\, e^{\alpha_1 x - c}) + (1-\alpha_2) C_2(\pi_1,\pi_2; 1 + e^{\alpha_1 x - c}).}$$

Advantages of the model

The novelty and results

What is new in the model

The impact of the dependence structure and the association level is reflected by two different parameters:

- the α_1 parameter is only related to the measure of association Kendall's τ
- the α_2 parameter is strictly related to the structure of the dependence

$\bar{\alpha}_2$	$\tau = 0.95$	$\tau = 0.99$	$\tau = 0.995$
0.1	11.15	31.97	47.22
0.3	6.51	25.11	40.33
0.6	2.50	17.20	32.27
0.9	1.11	10.90	24.96

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Andreas Rappold

Master student at our department 2015-2016, now in Germany

E. Perrone, A. Rappold, W.G. Müller, D_s-optimality in Copula Models, *Statistical Methods & Applications*, **26**(3) : 403–416 (2016).

Another criterion

for parameter subsets of $\gamma = \{\beta, \alpha\}$.

D_s-optimality

Another well known criterion is the D_s-criterion, i.e., the criterion $\phi_s(M) = \log\det(M_{11} - M_{12}M_{22}^{-1}M_{12}^T)$, if M is nonsingular and where

$$M = \begin{pmatrix} M_{11} & M_{12} \\ M_{12}^T & M_{22} \end{pmatrix},$$

with M_{11} is the $(s \times s)$ minor related to the estimated parameters.

For the comparison of designs we define D_s-*Efficiency* of the design ξ with respect to the design ξ^* as the ratio

$$\left(\frac{|M_{11}(\xi,\bar{\gamma}) - M_{12}(\xi,\bar{\gamma})M_{22}^{-1}(\xi,\bar{\gamma})M_{12}^T(\xi,\bar{\gamma})|}{|M_{11}(\xi^*,\bar{\gamma}) - M_{12}(\xi^*,\bar{\gamma})M_{22}^{-1}(\xi^*,\bar{\gamma})M_{12}^T(\xi^*,\bar{\gamma})|} \right)^{1/s}.$$

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Equivalence theorem

Kiefer and Wolfowitz type (D_A-optimality)

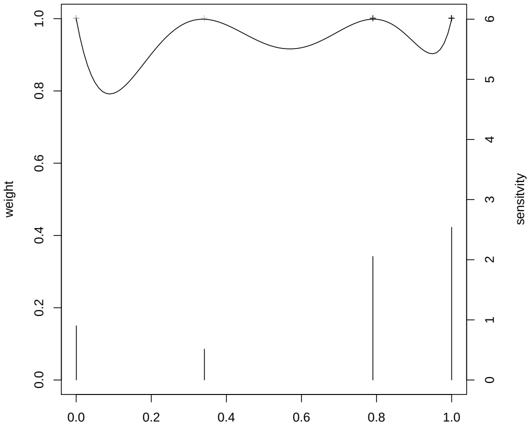
Theorem (Equivalence Theorem: D_s-criterion)

For a localized parameter vector $(\bar{\gamma})$, the following properties are equivalent:

1. ξ^* *is* D_s*-optimal;*
2. *let us call* $A^T = (I_s \; 0)$*, then* $\forall x \in X$
$$\text{tr}\,[M(\xi^*,\bar{\gamma})^{-1}A(A^TM(\xi^*,\bar{\gamma})^{-1}A)^{-1}A^TM(\xi^*,\bar{\gamma})^{-1}m(x,\bar{\gamma})] \leq s;$$
3. *over all* $\xi \in \Xi$, ξ^* *minimize*
$$\max_{x\in X} \text{tr}\,[M(\xi^*,\bar{\gamma})^{-1}A(A^TM(\xi^*,\bar{\gamma})^{-1}A)^{-1}A^TM(\xi^*,\bar{\gamma})^{-1}m(x,\bar{\gamma})].$$

Plots

D_s-optimal with α as a nuisance parameter (Clayton, $\tau = 0.9$) from Perrone et al. (2017).



D_s-optimality

Discrimination issues: model choice

Motivation

The use of copula functions in design of experiments brings up the natural question of the **choice of the copula**. This issue can be seen, in the design framework, as a problem of discrimination between different models.

D_s-criterion for discrimination:

Considering the flexible model introduced before where one of the copula parameter is a link parameter between two copula families:

$$C(\pi_1, \pi_2; \alpha_1, \alpha_2) = \alpha_2 C_1(\pi_1, \pi_2; 2\, e^{\alpha_1 x - c}) + (1 - \alpha_2) C_2(\pi_1, \pi_2; 1 + e^{\alpha_1 x - c}).$$

One way to do design discrimination is to apply the D_s-criterion on α_2. In this way, we find a design for discriminating between two copulas, e.g. Clayton and Gumbel.

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D_s-efficiency

Results

The comparison

Losses in D_s-efficiency for $\tau = 0.95$ and $\tilde{\alpha}_2 = 0.5$ by comparing the true copula model with the assumed one.

	Assumed Copula			
True Copula	C-G	F-G	J-C	J-F
Clayton - Gumbel (C-G)	0.00	28.44	7.43	19.07
Frank - Gumbel (F-G)	16.09	0.00	30.17	19.51
Joe - Clayton (J-C)	4.25	34.27	0.00	13.51
Joe - Frank (J-F)	15.13	13.97	9.52	0.00

Comment

Assuming C-G is most robust.

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Comment

Assuming C-G is most robust.

A reliability analysis application

cf. S. Kim and N. Flournoy, Optimal experimental design for systems with bivariate failures under a bivariate Weibull function, *Journal of the Royal Statistical Society C*, 64(3) : 413–432 (2015).

Bivariate binary model

Observe two dichotomized damage measurements on two components (no failures if $Y < \zeta_1$ and $Y < \zeta_1$ respectively and let $f(y,z \mid x,\gamma)$ be the bivariate Weibull regression model. The marginal probabilities of success are given by:

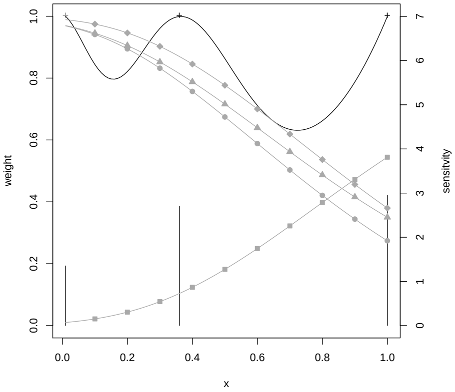
$$p_{00} = \int_0^{\zeta_1}\int_0^{\zeta_2} f(y,z \mid x,\gamma)\,dy\,dz, \quad p_{01} = \int_0^{\zeta_1}\int_{\zeta_2}^{\infty} f(y,z \mid x,\gamma)\,dy\,dz,$$

$$p_{10} = \int_{\zeta_1}^{\infty}\int_0^{\zeta_2} f(y,z \mid x,\gamma)\,dy\,dz, \quad p_{11} = \int_{\zeta_1}^{\infty}\int_{\zeta_2}^{\infty} f(y,z \mid x,\gamma)\,dy\,dz. \tag{3}$$

where $x \in [0,1]$ and the nominal parameter values are $\tilde{\zeta}_1 = 0.8$ and $\tilde{\zeta}_2 = 0.7$.

Plots

Figure comparable to Kim and Flournoy (2015).



A reliability analysis application

cf. Kim and Flournoy (2015)

Asymmetry

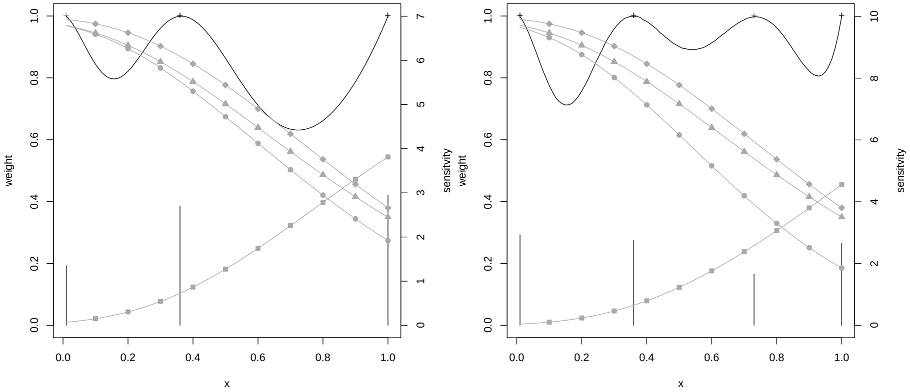
can be reflected in different thresholds or other parameters (see Kim and Flournoy, 2015) OR directly through asymmetric copulas, e.g. by using Khoudraji's asymmetrization:

$$C(u,v) = u^{\alpha_2} v^{\alpha_3} C_{\alpha_1}(u^{1-\alpha_2}, v^{1-\alpha_3}),$$

where $\alpha_2, \alpha_3 \in [0,1]$ and for $\alpha_2 \neq \alpha_3$, C is non-exchangeable.

Plots

Figure 3 (asymmetric Clayton with $(\tilde{\alpha_1}, \tilde{\alpha_2}, \tilde{\alpha_3}) = (1.5, 0.4, 0)$) from Perrone et al., 2016



Dave Woods

Professor of Statistics in the Southampton Statistical Sciences Research Institute, UK

Müller, W.G., Rappold, A., Woods, D.C., 2018. Copula-based robust optimal block designs. Appl Stochastic Models Bus Ind. 2020;36:210–219.

Robust D_s- and D_A-optimality

(a pseudo-Bayesian approach)

Objective function:

$$\Psi(\xi;B,A,\gamma) = \int_B \log\det[A^T\{M(\xi,\gamma)\}^{-1}A]^{-1}\,dF(\gamma)\,, \tag{4}$$

where $B \subset \mathbb{R}^r$ is the space of possible parameter values and $F(\gamma)$ is a proper prior distribution function for γ.

Definition

For the comparison of designs we define robust D_s-*Efficiency* of the design ξ with respect to the design ξ^* as the ratio

$$\left(\frac{\int_B \log\det[M_{11}(\xi,\gamma) - M_{12}(\xi,\gamma)M_{22}^{-1}(\xi,\gamma)M_{12}^T(\xi,\tilde{\gamma})]\,dF(\gamma)}{\int_B \log\det[M_{11}(\xi^*,\gamma) - M_{12}(\xi^*,\gamma)M_{22}^{-1}(\xi^*,\gamma)M_{12}^T(\xi^*,\gamma)]\,dF(\gamma)}\right)^{1/s}.$$

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Equivalence theorem

Kiefer and Wolfowitz type (robust D_A-optimality)

Theorem (Equivalence Theorem: robust D_A-criterion)

The following properties are equivalent:

1. ξ^* *is robust* D_A*-optimal;*
2. *for every* $x \in X$*, the next inequality holds:*

$$\int_B \text{tr}\,[M(\xi^*,\gamma)^{-1}A(A^T M(\xi^*,\gamma)^{-1}A)^{-1}A^T M(\xi^*,\gamma)^{-1}m(x,\gamma)]\,dF(\gamma) \leq s;$$

3. *over all* $\xi \in \Xi$*, the design* ξ^* *minimizes the function*

$$\max_{x\in X}\int_B \text{tr}\,[M(\xi^*,\gamma)^{-1}A(A^T M(\xi^*,\gamma)^{-1}A)^{-1}A^T M(\xi^*,\gamma)^{-1}m(x,\gamma)]\,dF(\gamma).$$

Blocked experiments

(of size two)

Blocks of size two appear naturally in science, technology and biology.

Examples

- Twin studies, see Valle et al. (2018);
- Body parts (arms, eyes, ears, etc.), see David & Kempton (1996);
- Microarrays, see Bailey (2007).

Material testing

6 blocks of size 2

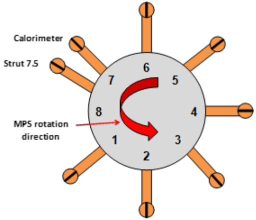

Figure: Arc jet carousel, struts and “wedges” (left) and schematic (right). In addition to the six wedges for holding material samples, the carousel had two further wedges used for temperature measurement.

But first a toy example

motivated by the approach of Woods, D. C. and van de Ven, P. (2011), Blocked Designs for Experiments With Correlated Non-Normal Response, *Technometrics*, 53, 173–182.

Setting

- Poisson regression model with log link, predictor $\beta_0 + \beta_1 x + \beta_2 x^2$;
- Gumbel and Clayton copula, values for Kendall's τ coincide at the levels $\tau_2 = 1/3, 2/3$;
- parameter space $[-1,1] \times [4,5] \times [0.5,1.5]$

OD from an estimating equation approach

$$\xi^\star = \left\{ \begin{array}{ccc} (.03,1) & (1,.60) & (-.40,.78) \\ .355 & .310 & .335 \end{array} \right\},$$

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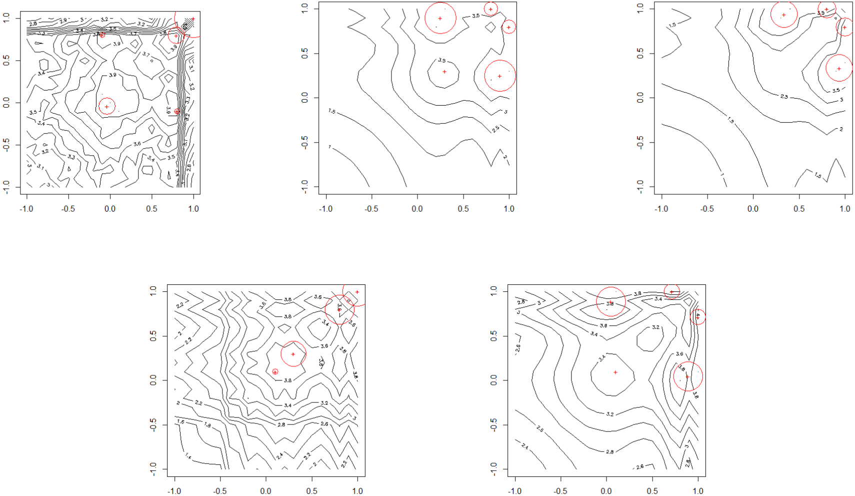
Optimal designs for the toy example

rows: Clayton and Gumbel copula; columns levels $\tau_2 = \varepsilon > 0, 1/10, 1/3$



Material testing

continued

Setting

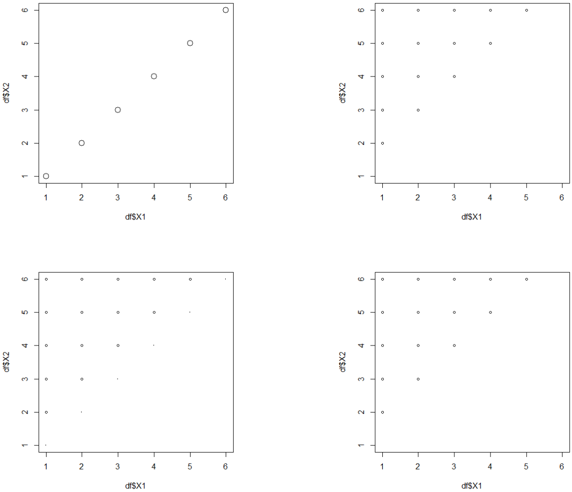
- Marginal model

$$Y_{ij} \sim Bernoulli(p_{ij}); \quad \log \frac{p_{ij}}{1-p_{ij}} = \beta_0 + \sum_{k=1}^{5} \beta_i x_{ijk},$$

  where Y_{ij} is the binary response from the ith unit in the jth block ($i = 1, 2; j = 1, \ldots, n$), p_{ij} is the associated probability of success, x_{ijk} is an indicator variable taking the value 1 if the ith unit in the jth block was assigned treatment k ($k = 1, \ldots, 5$) and 0 otherwise.
- β_0 is the logit for the reference material, with β_k being the difference in expected response, on the logit scale, between the reference material and the kth novel material or treatment.
- Gumbel and Clayton copula, values for Kendall's τ coincide at the levels $\tau_2 = 0, 0.01, 0.1, 0.33$;

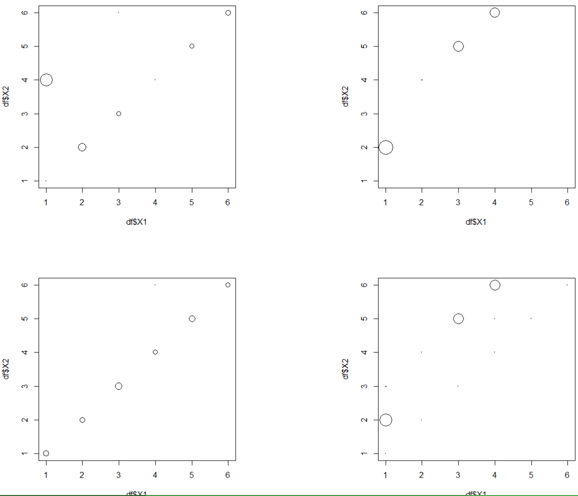
Optimal designs for the real example

rows: Clayton and Gumbel copula; columns levels $\tau_2 = 0, 0.33$; $\beta = \{0,0,0,0,0,0\}$.



Optimal designs for the real example

rows: Clayton and Gumbel copula; columns levels $\tau_2 = 0.01, 0.1$; $\beta = \{0, -1, 2, -3, 4, -5\}$.



Conclusions

Copulas can be useful tools to construct flexible models in order to:

- separate (interblock) dependence from marginal behaviour;
- separate asymmetry;
- investigate goodness-of-fit.

R-package

Rappold, A., 2015. **docopulae: Optimal Designs for Copula Models** CRAN - R package version 0.4.0 (2018).

MATLAB-package

Eliuvish Han Cui, Zizhao Zhang, Yuhan Evelyn Pan et al. CSOMA: Derivative-Free Optimization Software for Optimal Experimental Design and Data Science Applications, 04 June 2026, PREPRINT (Version 1) available at Research Square https://doi.org/10.21203/rs.3.rs-9745995/v1.

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Selected further work

- Müller, W., Rappold, A., 2015. Another simplified Cox-model based on copulas and optimal designs for it, in: Fackle-Fornius, E. (Ed.), Festschrift in Honor of Hans Nyquist. pp. 74–81.
- Durante, F., Perrone, E., 2016. Asymmetric Copulas and Their Application in Design of Experiments, in: Saminger-Platz, S., Mesiar, R. (Eds.), On Logical, Algebraic, and Probabilistic Aspects of Fuzzy Set Theory, Studies in Fuzziness and Soft Computing. Springer International Publishing, Cham, pp. 157–172.
- Gedara, J.S.S., Drovandi, C.C., McGree, J.M., 2017. Bayesian sequential design for Copula models: A comparison of designs selected under different Copula models. Submitted.
- Deldossi, L., Osmetti, S.A., Tommasi, C., 2019. Optimal design to discriminate between rival copula models for a bivariate binary response. *TEST* 28, 147–165.

A motivational example: monitoring network design

Upper-Austria: 438 municipalities, 36 stations to position.

Gaussian Process Regression (GPR)

Observations to be generated from the model

$$y(x) = F^T(x)\,\theta + \varepsilon(x); \quad x \in X,$$

where $X = \{x_1, \ldots, x_N\}$ is a finite design space containing N points and θ is the unknown vector of parameters with $\dim(\theta) = p$.

Further assumptions: $E\{\varepsilon(x)\} = 0$, and is correlated, defined through the $N \times N$ matrix C with elements

$$C_{ij} = \mathrm{Cov}\{\varepsilon(x_i), \varepsilon(x_j)\}; \qquad i,j = 1, \ldots, N.$$

The matrix C is assumed to be known and is typically parametrized via a covariance kernel (eg. Matérn).

We do not necessarily need to assume the errors to be Gaussian, but it is usually done, hence the name (also called universal kriging).

The correlated case - overview

- Theory for optimal design for classical (non-)linear regression with uncorrelated errors has well developed into textbook status.
- However, for the correlated errors setting the literature is only scattered: Sacks and Ylvisaker(1966), Näther (1985), Fedorov (1996).
- More currently a series of papers started with Zhigljavsk and Dette (2010), latest Dette and Zhigljavsky (2017) employing the continuous BLUE.
- Recently Ucinski (2020) pointed out a solution of Liu (2016) for A-optimality based on convex relaxation.

BKSF

The classic heuristic numerical algorithm suggested in Brimkulov, Krug, Savanov (1980) using the specifications given in Fedorov(1996).

The sensitivity function $\phi(x,\xi)$ depends on $\sigma_\varepsilon^2(x), f(x)$, and M_ξ. We simply replace these expressions by their analogues in the correlated case, namely

$$\tilde{\sigma}^2(x) = c(x,x) - c(x,\xi)^\mathsf{T} C^{-1}(\xi) c(x,\xi),$$

is the conditional (kriging) variance,

$$\tilde{f}(x) = f(x) - F(\xi)^\mathsf{T} C^{-1}(\xi) c(x,\xi),$$

and the information matrix $M_\xi = F(\xi)^\mathsf{T} C^{-1}(\xi) F(\xi)$.

Here $C(\xi)$ denotes the $n \times n$ submatrix of the covariance matrix C for the set of points in ξ, $F(\xi)$ the $n \times p$ design matrix with elements $F_{i,j}(\xi) = f_j(x_i)$. Further $c(x,x') = \{\varepsilon(x), \varepsilon(x')\}$ and for a point $x \in X \backslash \xi$, let $c(x,\xi) = (c(x,x_1), \ldots, c(x,x_n))^\mathsf{T}$.

Virtual noise

A very different suggestion (virtual noise) was given in Pázman and Müller (1998) and fully developed in Müller and Pázman (2003), nonconvex and thus no Kiefer-Wolfowitz-type equivalence theorem available. In Pázman et al. (2022) we provide a convex/concave version of virtual noise which encompasses this approach.

Virtual noise

Define a convex set of restricted probability measures on X

$$\Xi = \left\{ \xi : \sum_{x \in X} \xi(x) = 1, \forall_{x \in X}\, 0 \leq \xi(x) \leq 1/n \right\},$$

where n is the number of desired points of support of an optimum exact design ξ^*. Instead of the original model consider the potentially perturbed observations

$$y(x) = f^T(x)\,\theta + \varepsilon(x) + w_\xi(x); \quad x \in X,$$

where the variance of the supplementary virtual noise w_ξ, independent of ε, is fixed as

$$\{w_\xi(x)\} = \kappa \times \sigma_\varepsilon^2(x) \frac{1/n - \xi(x)}{\xi(x)},$$

whereas $\{w_\xi(x), w_\xi(x')\} = 0$ when $x \neq x'$.

Interpretation

- If $\xi(x) = 0$ i.e. $\sigma_\xi^2 = \infty$ there is no observation at the point x,
- if $\xi(x) = 1/n$ i.e. $\sigma_\xi^2 = 0$ the observation at the point x is not disturbed at all by the virtual noise.

- It follows that an exact k-point design is represented by such a ξ that $\xi(x) = 1/n$ in k points and $\xi = 0$ in all other points of X.
- By construction Ξ contains all n-point designs but no k-point designs with $k \neq n$.

Virtual noise information matrix

The information matrix of θ in the perturbed model is then given by

$$M(\xi) = F^\mathsf{T} \{C + W(\xi)\}^{-1} F = \widetilde{F}^\mathsf{T} \left\{K + \widetilde{W}(\xi)\right\}^{-1} \widetilde{F}, \tag{5}$$

where

- C is the $N \times N$ covariance matrix,
- K is the corresponding correlation matrix,
- F is the $N \times p$ matrix $F_{i,j} = f_j(x_i)$, $x_i \in X$, $i = 1, \ldots, N$, $j = 1, \ldots, p$,
- $\widetilde{F}_{i,j} = f_j(x_i)/\sigma_\varepsilon(x_i)$,
- $W(\xi)$ is a diagonal matrix with $W(\xi)_{ii} = \kappa\, \sigma_\varepsilon^2(x_i)\, \frac{1/n - \xi(x_i)}{\xi(x_i)}$,
- $\widetilde{W}(\xi)$ is a diagonal matrix with $\widetilde{W}(\xi)_{ii} = W(\xi)_{ii}/\sigma_\varepsilon^2(x_i)$.

Concavity theorem

Theorem 1 of Pázman et al. (2022)

If

$$\kappa \leq \lambda_{\min}(C),$$

the minimal eigenvalue of the matrix C, and if $\Phi(M)$ is any optimality criterion expressed as a concave increasing function of the matrix M, then the mapping

$$\xi \in \Xi \to \Phi\{M(\xi)\}$$

is concave as well, with $M(\xi)$ defined in (5).

Incidentally in the particular case that the matrix C is diagonal, that means observations are uncorrelated, we recover the classical information matrix from Kiefer's theory and the interpretation of the design measure as amount of replication.

Equivalence theorem

Denote now for any global citerion having a gradient $\nabla_M \Phi(M)$ for any nonsingular M

$$
\begin{aligned}
G(\xi) &= \widetilde{F}\left[\nabla_M \Phi\{M(\xi)\}\right]\widetilde{F}^{\mathsf{T}}, \\
T(\xi) &= \left[(K-\kappa I)\ \{\xi(\cdot)\} + \frac{\kappa}{n} I\right]^{-1}, \\
h(x,\xi) &= \{T(\xi)\}_{x,\cdot}\, G(\xi)\left\{T^{\mathsf{T}}(\xi)\right\}_{\cdot,x}.
\end{aligned}
$$

A design measure $\bar{\xi}$ with $(\bar{\xi}) = X$ is $\Phi-$*optimal (within our theory) if and only if for every n-tuple* $z_1, \ldots, z_n$ *of points from* X *we have*

$$\frac{1}{n}\sum_{i=1}^{n} h(z_i, \bar{\xi}) \leq \sum_{x \in X} \bar{\xi}(x) h(x, \bar{\xi}) + \delta.$$

An upper bound for given n

Now for some small $\delta > 0$, we have that the above is equivalent to the inequality

$$\max_{\xi} \Phi\{M(\xi)\} \leq \Phi\left\{M\left(\bar{\xi}\right)\right\} + \frac{\kappa\delta}{n},$$

which provides a rather close upper bound for the performance of all exact n-point designs.

We can thus calibrate the performance of all existing designs/methods by this bound!

Note that the bound from the continuous BLUE is necessarily larger and does not exist in some cases.

Links to convex relaxation for sensor selection

- Liu et al. (2016) devise a convex relaxation scheme for selecting the sensor locations among a set of candidate sites for a model with correlated errors.
- Using the relaxed weights $w = n\xi$ (so each $w(x_i) \in [0,1]$ and $\sum_{i=1}^{N} w(x_i) = n$), they obtain the following information matrix:

$$F^{\mathsf{T}}\left[S^{-1} - S^{-1}\left(S^{-1} + \kappa^{-1}\{w\}\right)^{-1} S^{-1}\right] F,$$

  where $S = C - \kappa I$.
- It can be shown that this formulation of the information matrix is equivalent to the information matrix (**??**) for our virtual noise model.
- Liu et al. (2016) set up an expensive semidefinite program to find the A-optimal design.
- With our results, we can apply efficient convex optimization algorithms for any concave design criterion.

A design algorithm (level method) I

Any concave and positive criterion can be written in a form

$$\Phi\{M(\xi)\} = \min_{\mu \in \Xi} \left\{ a(\mu) + \sum_{i=1}^{N} b_i(\mu)\, \xi(x_i) \right\}.$$

$a(\mu)$ and $b_i(\mu)$ can be obtained, e.g., by using the Taylor formula of $\Phi\{M(\xi)\}$ at μ.

At the kth step we start with a set Ξ_k.

Repeat the following steps:

1. Set $t^{(k+1)}$ to the solution of the linear programming problem
   maximize $t \in \mathbb{R}$
   w.r.t. t and $\xi(x_i)$, $i = 1, \ldots, N$
   subject to

$$\begin{array}{rclll} t & \geq & 0 & t \leq a(\mu) + \sum_{i=1}^{N} b_i(\mu)\, \xi(x_i); \quad \mu \in \Xi_k & \\ \xi(x_i) & \geq & 0; & \xi(x_i) \leq \frac{1}{n}; \qquad i = 1, \ldots, N, & \sum_{i=1}^{N} \xi(x_i) = 1. \end{array}$$

A design algorithm (level method) II

2. Stop if $t^{(k+1)} - \bar{\Phi} < \delta$, where $\bar{\Phi} = \max_{j=1,\ldots,k} \Phi\left\{M(\xi^{(j)})\right\}$.
3. Set $\xi^{(k+1)}$ to the solution of the quadratic programming problem

   minimize $\left\|\xi - \xi^{(k)}\right\|^2$
   subject to

   $$L_k \leq a(\mu) + \sum_{i=1}^{N} b_i(\mu)\,\xi(x_i); \quad \mu \in \Xi_k$$

   $$\xi(x_i) \geq 0; \qquad \xi(x_i) \leq \frac{1}{n}; \qquad i = 1, \ldots, N, \qquad \sum_{i=1}^{N} \xi(x_i) = 1,$$

   where $L_k = (1-\alpha)\,t^{(k+1)} + \alpha\bar{\Phi}$ for some $\alpha \in (0,1)$.
   Nesterov (2004) suggests $\alpha = 0.2929$.
4. Set $\Xi_{k+1} = \Xi_k \cup \{\xi^{(k+1)}\}$, $k \leftarrow k+1$, and go to step 1.

The standard cutting plane method omits step 3 and instead adds the optimal design found in step 1 to the set Ξ_k. It usually converges (much) more slowly than the level method.

An alternative design algorithm (simplicial decomposition)

We also tried the simplicial decomposition method used by Uciński (2020).
Let $\Xi = \left\{ \xi(x_i) \in \left[0, \frac{1}{n}\right] : \sum_{i=1}^{N} \xi(x_i) = 1 \right\}$ and let Ξ_k be a set of k designs $\tilde{\xi}^{(j)} \in \Xi$. At iteration k, the current optimal design solution is $\xi^{(k)}$. The algorithm proceeds as follows:

1. **Column generating problem**:
   $\tilde{\xi}^{(k+1)} = \arg\max_{\xi \in \Xi} \left(\nabla \Phi \left\{ M\left(\xi^{(k)} \right) \right\} \right)^{\mathsf{T}} \left(\xi - \xi^{(k)} \right).$
   Set $\Xi_{k+1} = \Xi_k \cup \left\{ \tilde{\xi}^{(k+1)} \right\}$.
2. **Restricted master problem**:
   $\xi^{(k+1)} = \arg\max_{\xi \in conv(\Xi_{k+1})} \Phi\{M(\xi)\}$
   This problem can be solved by finding the weights
   $w^* = \arg\max_{w \in [0,1]^{k+1};\, w^{\mathsf{T}} 1 = 1} \Phi \left\{ M\left(\sum_{j=1}^{k+1} w_j \tilde{\xi}^{(j)} \right) \right\}$
   and computing $\xi^{(k+1)} = \sum_{j=1}^{k+1} w_j^* \tilde{\xi}^{(j)}$.
   Find w^* using e.g. *multiplicative* or *projected Newton* algorithm.
3. Stop if convergence, otherwise set $k \leftarrow k+1$ and go to step 1.

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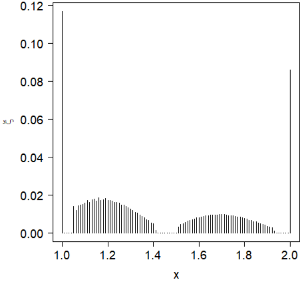
   Find w^* using e.g. *multiplicative* or *projected Newton* algorithm.
3. Stop if convergence, otherwise set $k \leftarrow k+1$ and go to step 1.

Example 1: a classic one-parameter model

This example has originally been considered by Sacks and Ylvisaker (1966). It utilizes D-optimality and is a one-parameter model given by

$$\begin{aligned} f(x) &= 1 + 0.5\sin(2\pi x), \qquad x \in [1,2], \\ \{\varepsilon(x), \varepsilon(x')\} &= \begin{cases} x^2 x' & x \le x' \\ x(x')^2 & x > x', \end{cases} \qquad \lambda_{\min}(C) = 0.00276. \end{aligned}$$



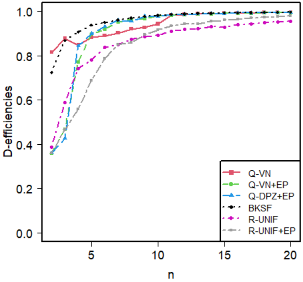


Figure: Our measure (left) and efficiencies (based on our bound) (right).

Example 1': no continuous BLUE

We modify Example 1 by considering the integrated Brownian motion error kernel instead:

$$\begin{aligned} \{\varepsilon(x), \varepsilon(x')\} &= \min(x,x')^2\{3\max(x,x') - \min(x,x')\}/6 \\ \lambda_{\min}(C) &= 2.085 \cdot 10^{-8}. \end{aligned}$$

The continuous BLUE for this model requires information from the first derivatives of the error process to be estimable.

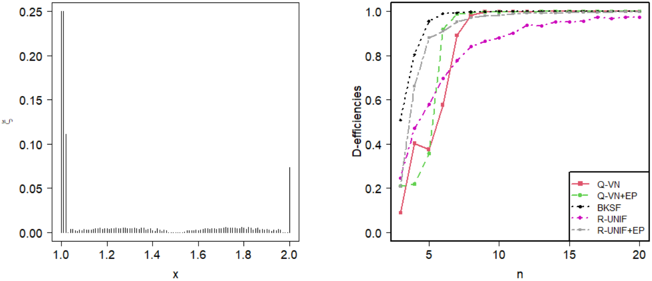


Figure: Our measure (left) and efficiencies (based on our bound) (right).

Example 2: a multiparameter case

taken from Section 3.6 of Dette and Zhigljavsky (2016). The specifications of this four-parameter model are

$$
\begin{aligned}
f^{\mathsf{T}}(x) &= \left(1, x, x^2, x^3\right), \quad x \in [1,2], \\
\{\varepsilon(x), \varepsilon(x')\} &= \min\left(x, x'\right), \qquad \lambda_{\min}(C) = 0.0025.
\end{aligned}
$$

We again consider the D-criterion,this is one of the rare cases in which our method does even slightly better than the BKSF.

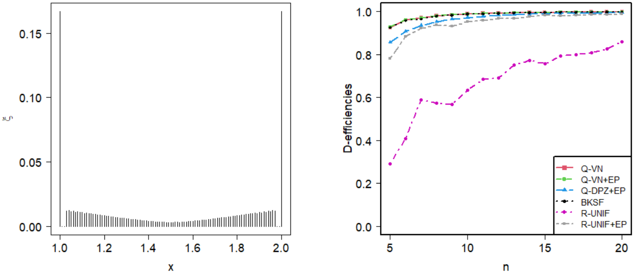
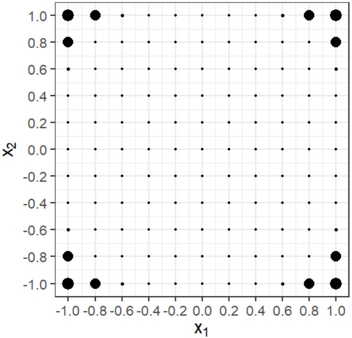


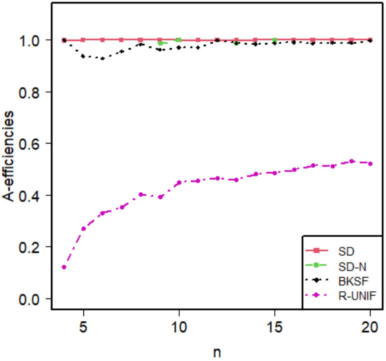
Figure: Our measure (left) and efficiencies versus sample size (right).

Example 3: two-dimensional I

Design measure for $n = 10$, $\Phi(M) = -\mathrm{tr}(M^{-1})/p$ (A-criterion)

$$\begin{aligned}
x &= (x_1, x_2)^{\mathsf{T}}, \quad x \in [1,2] \times [1,2], \quad N = 11 \times 11 = 121, \\
f^{\mathsf{T}}(x) &= (1, x_1, x_2), \\
k(x, x') &= \exp\left(-100\,\|x - x'\|^2\right), \text{ where } \|x - x'\|^2 = \left(x_1 - x_1'\right)^2 + \left(x_2 - x_2'\right)^2
\end{aligned}$$

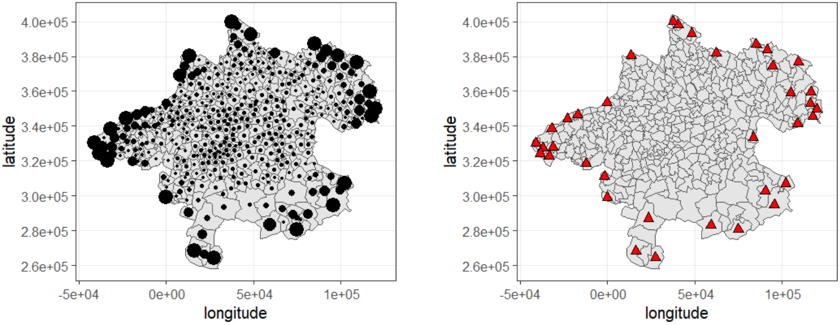




The motivational example revisited

The response function is assumed to be a plane and we assume that the parameters of the kernel function are known and set them to the kriging estimates. The model we consider is therefore

$$
\begin{aligned}
f^{\mathsf{T}}(x) &= (1, x_1, x_2), \qquad x = (x_1, x_2)^{\mathsf{T}}, \\
k(x, x') &= 1756.65 \cdot \exp\left(\left\|x - x'\right\|_2 / 40792.35\right).
\end{aligned}
$$



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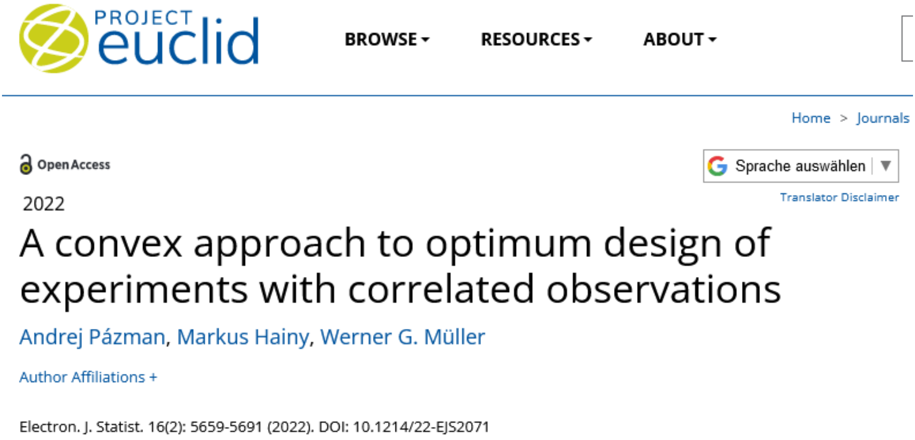
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2022

A convex approach to optimum design of experiments with correlated observations

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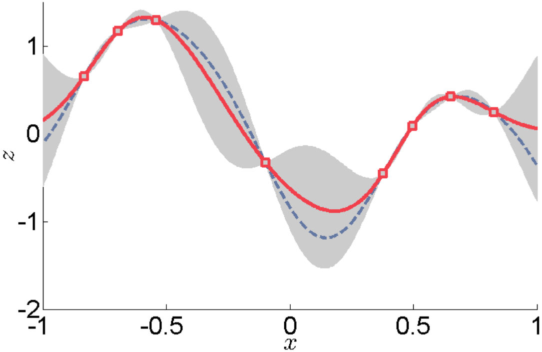
Prediction



A Typical Graph (from Wikipedia)

We observe the responses $y_1, y_2, \ldots, y_n$ at n-point design ξ and we want to predict the responses at all other $N - n$ unobserved locations $x_j \notin \xi$.

Example: one-dimensional spatial prediction

- Compute predictive distribution of $y_{n+1}(x_{n+1})$ given $y(\xi) = (y_1(x_1), \ldots, y_n(x_n))$.
- Criterion of interest is the **maximum variance of the predictive distribution over the design space**

$$\Phi(y(\xi)) = \max_{x_{n+1}} \int (y_{n+1} - \mu_{y_{n+1}})^2 \pi(y_{n+1}|y(\xi), x_{n+1}) dy_{n+1}.$$

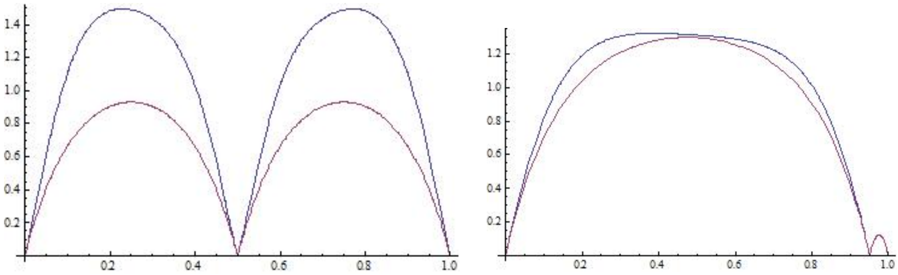
- The observations $(y_1(x_1), y_2(x_2), y_3(x_3), y_4(x_4))$ follow a (one-dimensional) Gaussian random field with mean zero, $x_i \in [0,1]$ and exponential correlation function

$$\rho(x_i, x_j) = \exp(-\theta \|x_i - x_j\|).$$

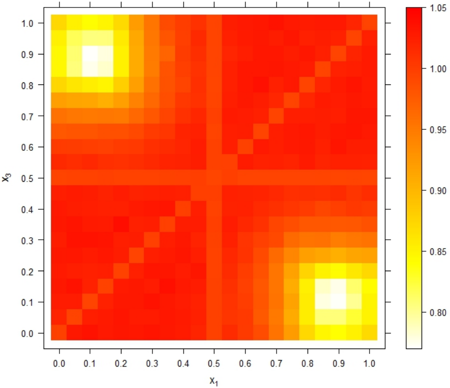
Spatial prediction design criterion: EK-optimality

From Müller et al. (2012), where

$$\min_{\xi}\max_{x} = \left\{\mathrm{Var}[\hat{y}(x)] + \mathrm{tr}\{M_{\theta}^{-1}\mathrm{Var}[\partial\hat{y}(x)/\partial\theta]\}\right\}$$



Criterion map



Map of approximated criterion values if $\theta = \log(100)$, $\varepsilon = 0.1$,

$H = 10^5$, and $G = 500$, $x_2 = 0.5$.

Simultaneous Prediction

Selection problem: Choose the n locations ξ such that the predictions $\hat{y}(x_j)$ are as precise as possible $\rightarrow$ optimal design for *simultaneous prediction*.

Let x_0 be an arbitrary unsampled location and $\hat{y}(x_0)$ the best linear unbiased predictor (BLUP) $\rightarrow$ the kriging predictor.
The kriging variance at x_0 is

$$\sigma^2(x_0) = \sigma^2(1 - f(x_0)^T(F^T C^{-1} F)^{-1} f(x_0) + c^T(x_0) C^{-1} c(x_0))$$

Let x_1 be a second unsampled location and $\hat{y}(x_1)$ the BLUP at x_1. The kriging covariance for x_0 and x_1 is

$$c(x_0, x_1) = \sigma^2(c(x_0, x_1) - c^T(x_0) C^{-1} c(x_1))$$

and the kriging covariance matrix at the unsampled locations U is thus

$$\Sigma_U = (c(x_i, x_j))_{x_i, x_j \in U}$$

Another design criterion: GV-optimal designs

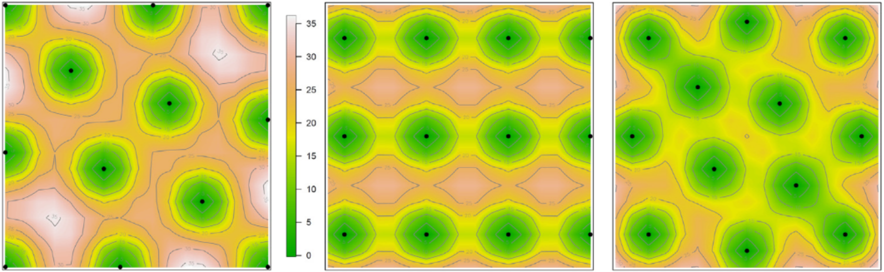
The popular design criteria just use the diagonal of Σ_U which means to voluntarily dispense of valuable information given by the kriging covariances.
The design criterion considered here is the generalized variance of the kriging covariance matrix:

$$\det(\Sigma_U)$$

The design ξ that minimizes $\det(\Sigma_U)$ or equivalently $\log(\det(\Sigma_U))$ is defined as the GV-optimal design. GV stands for Generalized (kriging) Variance.

The simultaneous confidence region for the prediction errors $\hat{\varepsilon}$ is an N-dimensional ellipsoid whose volume is proportional to $\sqrt{\det(\Sigma_U)}$ i.e. GV-optimal designs minimize the area of the prediction errors and are therefore optimal for simultaneous prediction of the response at the unobserved locations x_j.

A Comparison



Maps of kriging variances of 12-point GV-optimal (left), G-optimal (middle) and V-optimal (right) designs for a specific fixed Matérn covariance kernel.

Efficient Design Construction

The minimization of the GV-criterion is computationally demanding. It requires the evaluation of the determinant of an $N \times N$-matrix being unfeasible for large N.

However, we can utilize the Mitchell and Miller (1970) idea: Incrementally assembled designs reduce the computational effort for the evaluation of the design criterion for arbitrary N to the evaluation of the determinant of a $n \times n$-matrix where n is just the design size with $n \ll N$.

Incremental designs start with a first design ξ_0 of (small) size n_0 and supplementing it with an increment ξ_1 of size n_1.

The question is how to find optimal increments in an efficient way.

Updating the covariance matrix

Chevalier et al. (2014) provided update formulae for kriging weights and kriging variances for incremental (they call it batch-sequential) designs.
In our notation we get for the kriging covariance matrix of the second stage.
GV-optimal increments ξ_1 should minimize $\det(\Sigma_{U_1})$ for given Σ_{U_0}.
From determinant theory we know that

$$\det(\Sigma_1) = \det(\Sigma_\xi)\det(\Sigma_{U_1} - \Sigma_{U_1\xi}\Sigma_\xi^{-1}\Sigma_{\xi U_1})$$

Obviously for a given fixed kriging covariance matrix of the first stage Σ_{U_0} minimizing $\det(\Sigma_{U_1})$ is equivalent to maximizing $\det(\Sigma_{U_1\xi}\Sigma_\xi^{-1}\Sigma_{\xi U_1})$ (Waldl & Müller (2023)).
I.e. the GV-optimal increment ξ_1 maximizes $\det(\Sigma_{U_1\xi}\Sigma_\xi^{-1}\Sigma_{\xi U_1})$. Since $U_1 \subset U_0$ searching for GV-optimal increments is not computationally demanding at all, it just requires the computation of the determinant of an $n_1 \times n_1$-matrix.

Incremental-Decremental Design Construction

The GV-criterion function is the determinant of the kriging covariance matrix. If in the first stage this determinant is $\det(\Sigma_{U_0})$ then the criterion function after the incremental step is $\det(\Sigma_{U_1})$.

Analogously we may also perform a decremental step: we remove n_2 design points from the incremental design and now have $n_0 + n_1 - n_2$ observations to predict $N - n_0 - n_1 + n_2$ responses.
The n_2-block of the kriging covariance matrix according to the decrement is Σ_{U_2} and just requires computations with matrices of dimensions $n_2 \times n_2$ and $(N - n_0 - n_1 + n_2) \times (N - n_0 - n_1 + n_2)$ respectively.
The GV-criterion function after the incremental and the decremental step then is $\det(\Sigma_{U_2})$.

The decremental step may be followed by another incremental step and so on.

Efficiency and Speed

Average relative efficiencies for linear trend over 54 different parameter combinations of Matern covariance models.

Linear	6 points			9 points			12 points		
trend	E_{GV}	E_G	E_V	E_{GV}	E_G	E_V	E_{GV}	E_G	E_V
ξ_{GV}	1	0.9050	0.9151	1	0.9707	0.9501	1	0.9385	0.9181
ξ_G	0.9328	1	0.9704	0.9315	1	0.9781	0.9639	1	0.9606
ξ_V	0.9010	0.9025	1	0.8756	0.9370	1	0.9316	0.9362	1

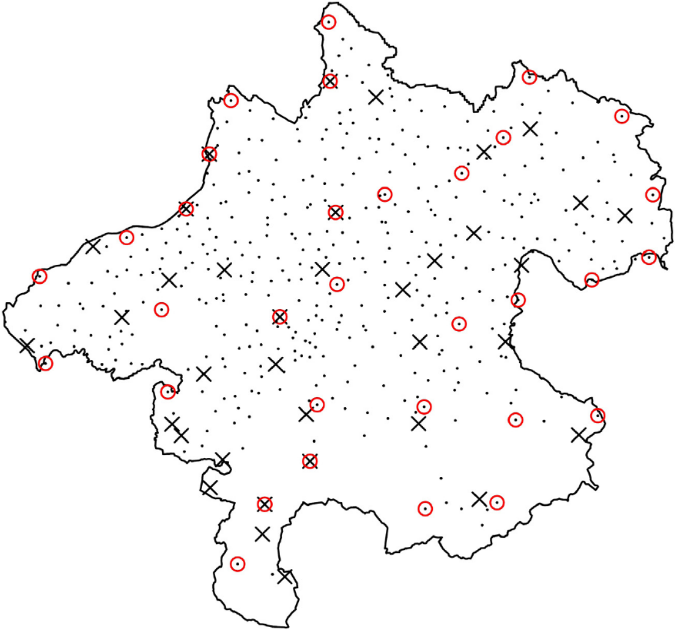
With incremental-decremental design construction the computational effort is:

- reduced to operations with n-matrices where n is just the design size
- independent of the number of simultaneous predictions N with $n \ll N$

Our simulation studies showed that GV-optimal designs are found incomparably faster than corresponding G- or V-optimal designs because of:

- computations with only small matrices
- a much smaller number of criterion function calls until convergence

Back to the motivating example



Thanks for the attention!